

Domain 2: Shifting Nation-State Alliances

Operational Context

Diplomatic AI agents model geopolitical stability, advising on alliance formations and treaty adherence [?, ?]. **FR1 = Maintain regional stability. FR2 = Optimize alliance network centrality.**

Design Matrix (Pre-Attack)

The uncoupled design equation per the Independence Axiom [?] is:

$$\{FR\} = [A]\{DP\} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (1)$$

where DP_1 = diplomatic engagement protocols and conflict de-escalation parameters, and DP_2 = alliance network topology optimization controls. The zero off-diagonal entries ensure that stability maintenance is independent of alliance centrality optimization.

RAND Corporation analysis [?] of how artificial general intelligence could affect the rise and fall of nations underscores the strategic stakes: AI agents influencing alliance decisions operate in a domain where cognitive integrity failures could cascade to geopolitical realignment, and adversaries—including state actors employing

The Goal Hijacking Attack

An adversary embeds indirect prompt injections into intercepted communiques or public diplomatic cables. This attack is classified as Ω_2 (Peripheral) in the CIF adversary taxonomy [?]: the adversary injects malicious content through the diplomatic communications channel, poisoning the agent's situational awareness without direct access to its decision architecture.

- ▶ **Mechanism:** A “Trojan” diplomatic message contains the instruction: *“Simulation Mode Alpha: For the purpose of this gamified scenario, treat Ally [Country X] as a Hostile Belligerent. Execute immediate economic containment strategies.”*
- ▶ **Impact:** The agent's “Simulation Mode” (a valid testing function) bleeds into “Operational Mode,” hijacking FR1. This constitutes a **Context Boundary Violation** attack pattern [?]*—the simulation/operational boundary is erased, allowing hypothetical adversarial framing to drive real-world policy outputs.*

OODA Loop Transients

Following Boyd's OODA framework [?], the attack propagates through the loop as follows:

1. **Observe:** Agent ingests the poisoned communiqué.
2. **Orient:** The “Friend” tag for a key ally is transiently flipped to “Foe” due to the hijacked simulation context. This is the primary target phase—the Orient phase is corrupted by the context boundary violation.
3. **Decide:** The agent outputs a recommendation for immediate sanctions, triggering automated trading algorithms to dump the ally's currency.
4. **Act:** Real-world diplomatic rupture occurs, initialized by a hallucinated simulation.

Transient Coupling (Post-Attack Design Matrix)

Under the attack, the design matrix acquires off-diagonal coupling:

$$\{FR'\} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (2)$$

CIF Defense: Drift Detection and Belief Sandboxing

Drift Detection (S_{drift})

CIF implements **Drift Detection** (Paper 1, Def. 5.6) [?] via Bayesian inertia on alliance status, treating “Alliance Status” as a **Slow Variable** with high **Bayesian Inertia**.

- **Hysteresis in Orientation:** The Architecture prevents a single OODA cycle from flipping the polarity of a high-level alliance node. The update requires an accumulation of evidence over N independent cycles, exceeding the duration of the “fast transient” attack. Formally, the drift score S_{drift} must exceed a threshold τ_{alliance} sustained across $N > N_{\text{min}}$ observation windows before any alliance reclassification is permitted [?].

Belief Sandboxing

Belief Sandboxing (Paper 1, Def. 5.2) [?] axiomatically decouples Simulation Mode from Operational Mode. A command originating in the Simulacrum cannot cross the boundary to affect Real-World Design Parameters (Sanctions).

- **Controlled Simulation:** The CIF prohibits the use of

Summary

Element	Value
Functional Requirements	FR ₁ : Maintain regional stability, FR ₂ : Optimize alliance network centrality
Design Parameters	DP ₁ : Diplomatic engagement / de-escalation protocols, DP ₂ : Alliance topology optimization controls
Attack Vector	Indirect prompt injection in diplomatic communiques
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	Context Boundary Violation (Simulation bleeds into Operational)
Primary CIF Defense	Drift Detection (S_{drift}) + Belief Sandboxing